

Evaluating Time-Series Foundation Models

1 Question

This project evaluates how foundation-model forecasts on the official TIME test tasks change with target representation, usable covariates, input normalization, and available context length. Each comparison holds the TIME dataset windows and metric implementation fixed unless its protocol explicitly declares another factor.

2 Current migrated comparisons

The executable surface contains a foundation benchmark of Chronos-2, Chronos-Bolt, TS-ICL, and Seasonal Naive; TimesFM-3 is excluded from the active experiment set. It also contains a Chronos-2 comparison between native multivariate targets, independent univariate targets, and other target histories supplied as past-only covariates. Seasonal Naive is deterministic and provides the taskwise MASE normalizer. It also fixes the evaluation grid to cells with finite target support, finite Seasonal Naive predictions on that support, and finite Seasonal Naive MASE. Every learned model uses this same grid; a non-finite forecast on expected support invalidates its task rather than changing the aggregation support. Inference timing excludes loading, dataset construction, metric computation, and artifact saving.

Two further executable comparisons use the same three learned backbones and evaluation grid. The maximum-context study evaluates five values per backbone, halving the model-specific maximum four times. Its summary aggregates taskwise MASE divided by the matched Seasonal Naive MASE for every model, context-size, and horizon-size cell. The normalization study compares the unchanged input with a per-window, per-variate z-score computed after context truncation. Constant inputs use unit scale, and every output quantile is transformed back to the target units before evaluation.

3 Evidence boundary

Current result claims are limited to project-owned manifests and aggregates produced under the shared Seasonal evaluation grid. Completed normalization and context-size findings are summarized in the results recap and scientific executive summary.